Socioeconomic indicators for various countries

Dependent variable (Y)

1.Gdpp

The GDP per capita. (Calculated as the total GDP divided by the total population)

Independent variables (X)

1.Export

Exports of goods and services per capita, (presented as a percentage of the GDP per capita)

2.child_mort

Death of children under 5 years of age per 1000 live births

3. Health

Total health spending per capita, (presented as a percentage of GDP per capita)

4. Imports

Imports of goods and services per capita. (Given as a % of the GDP per capita)

5. Income

Net income per person

6. Inflation

The measurement of the annual growth rate of the total GDP

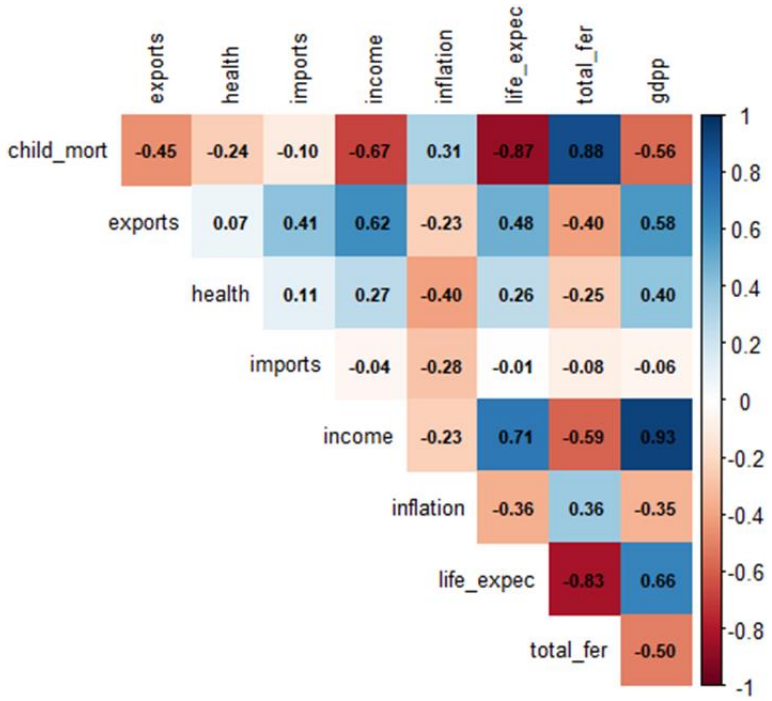
7. Life expect

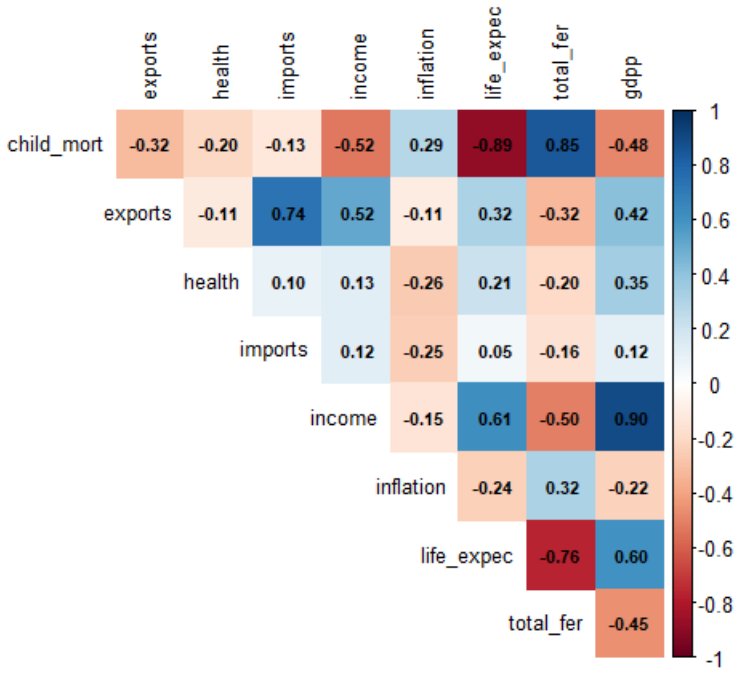
The average number of years a newborn child would live if current mortality patterns remain the same.

8.Total fer

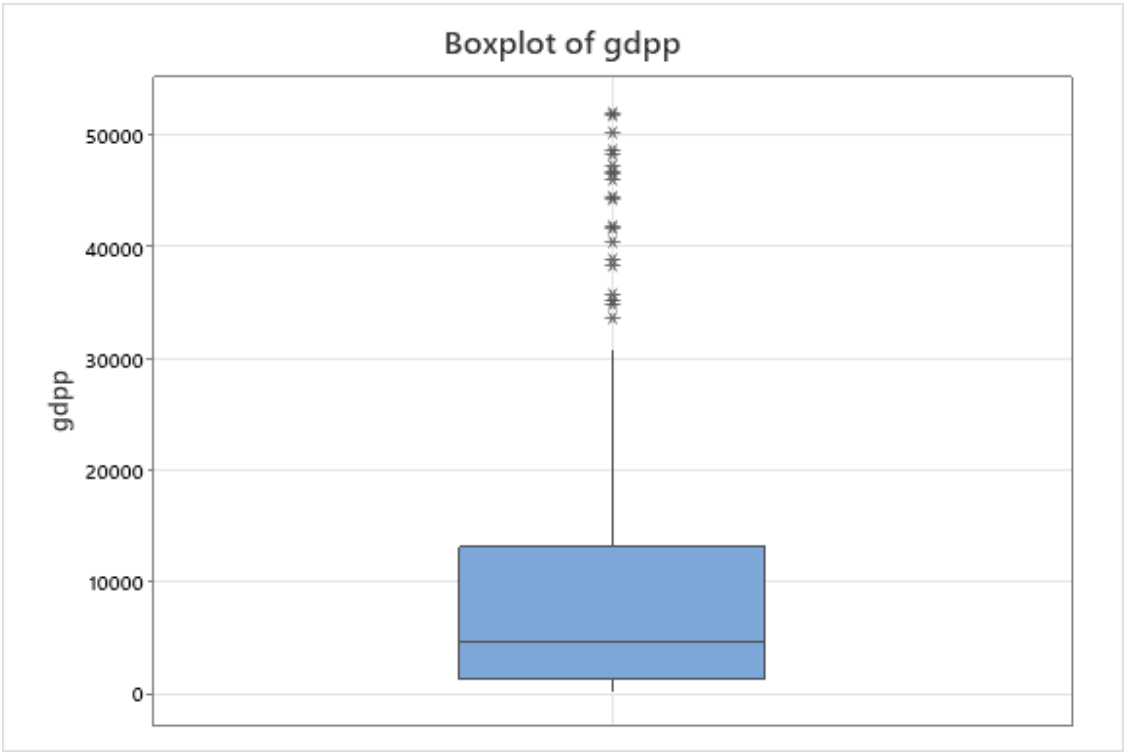
The number of children that would be born to each woman if current age-fertility rates remain the same.

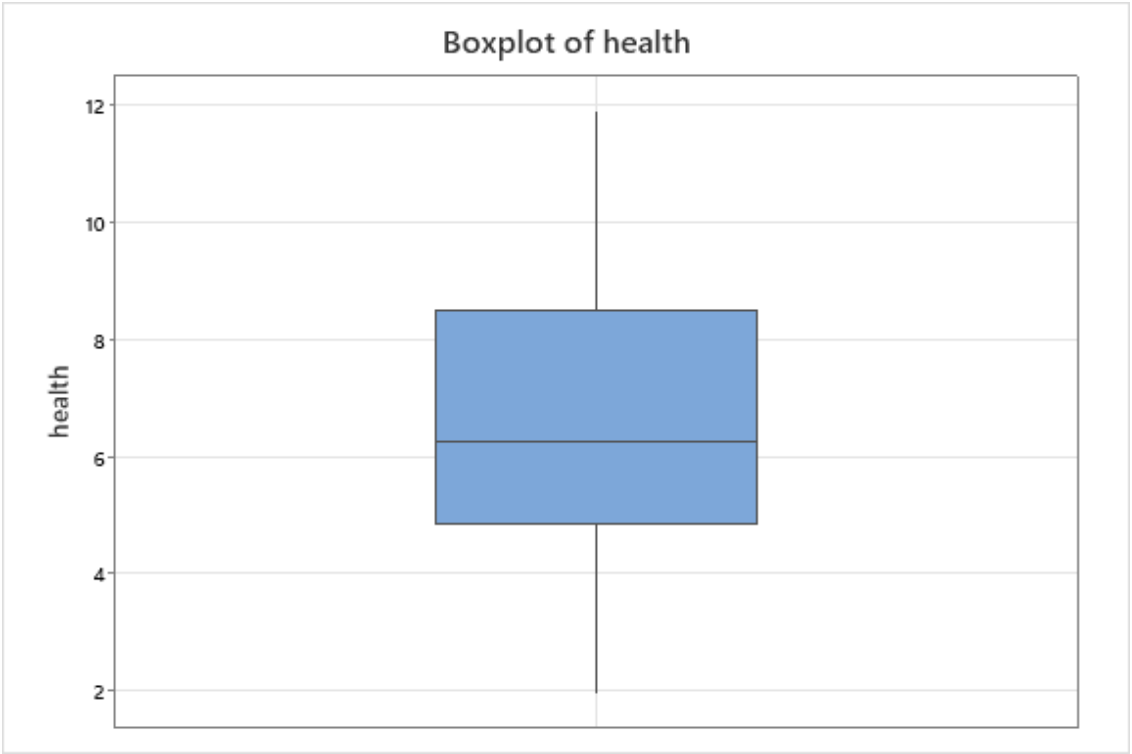
Corrplot

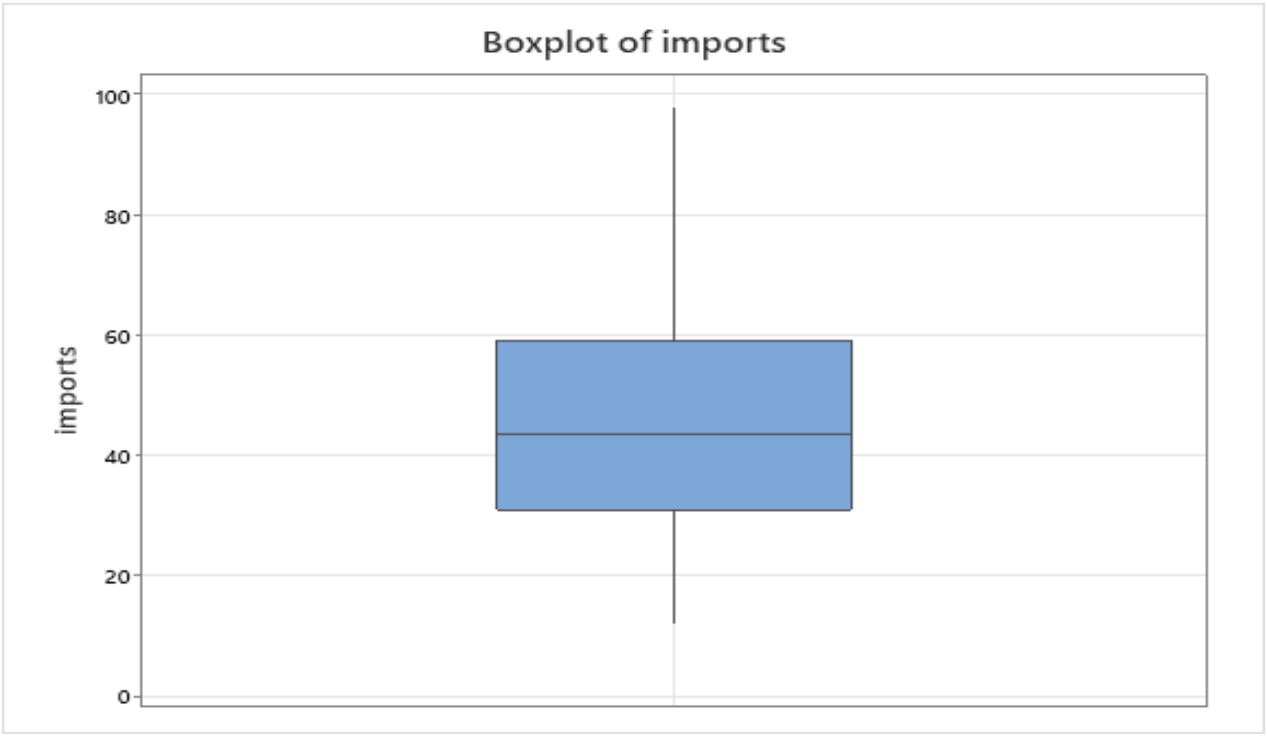

exports
health
imports
income
inflation
life_expec
total_fer
gdpp
child_mort -0.45 -0.24 -0.10 -0.67 0.31 -0.87 0.88 -0.56
exports 0.07 0.41 0.62 -0.23 0.48 -0.40 0.58
health 0.11 0.27 -0.40 0.26 -0.25 0.40
imports -0.04 -0.28 -0.01 -0.08 -0.06
income -0.23 0.71 -0.59 0.93
inflation -0.36 0.36 -0.35
life_expec -0.83 0.66
total_fer -0.50
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

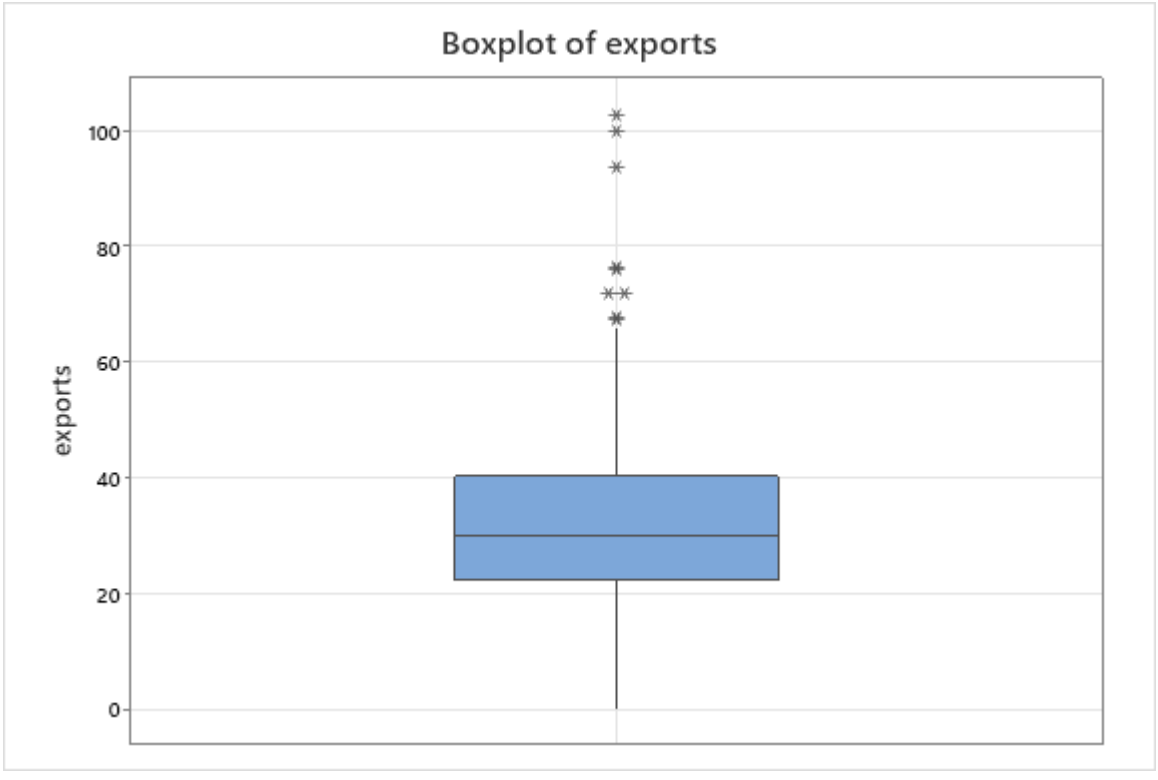

exports
health
imports
income
inflation
life_expec
total_fer
gdpp
child_mort -0.32 -0.20 -0.13 -0.52 0.29 -0.89 0.85 -0.48
exports -0.11 0.74 0.52 -0.11 0.32 -0.32 0.42
health 0.10 0.13 -0.26 0.21 -0.20 0.35
imports 0.12 -0.25 0.05 -0.16 0.12
income -0.15 0.61 -0.50 0.90
inflation -0.24 0.32 -0.22
life_expec -0.76 0.60
total_fer -0.45
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

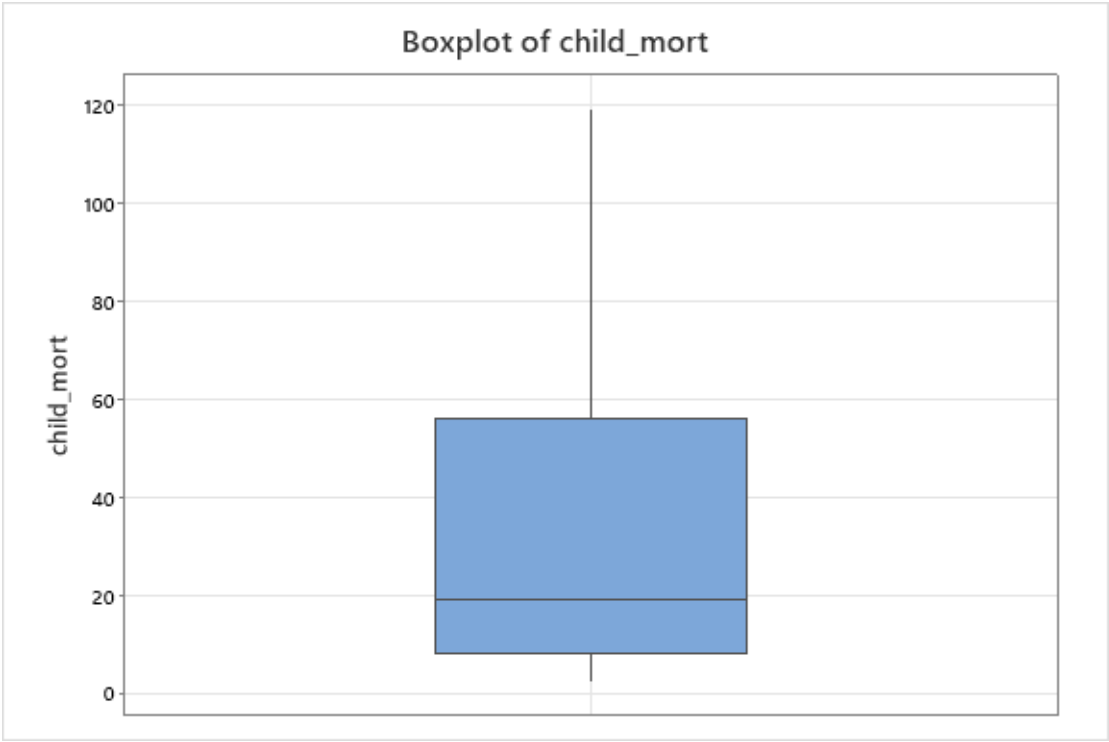
Box plots

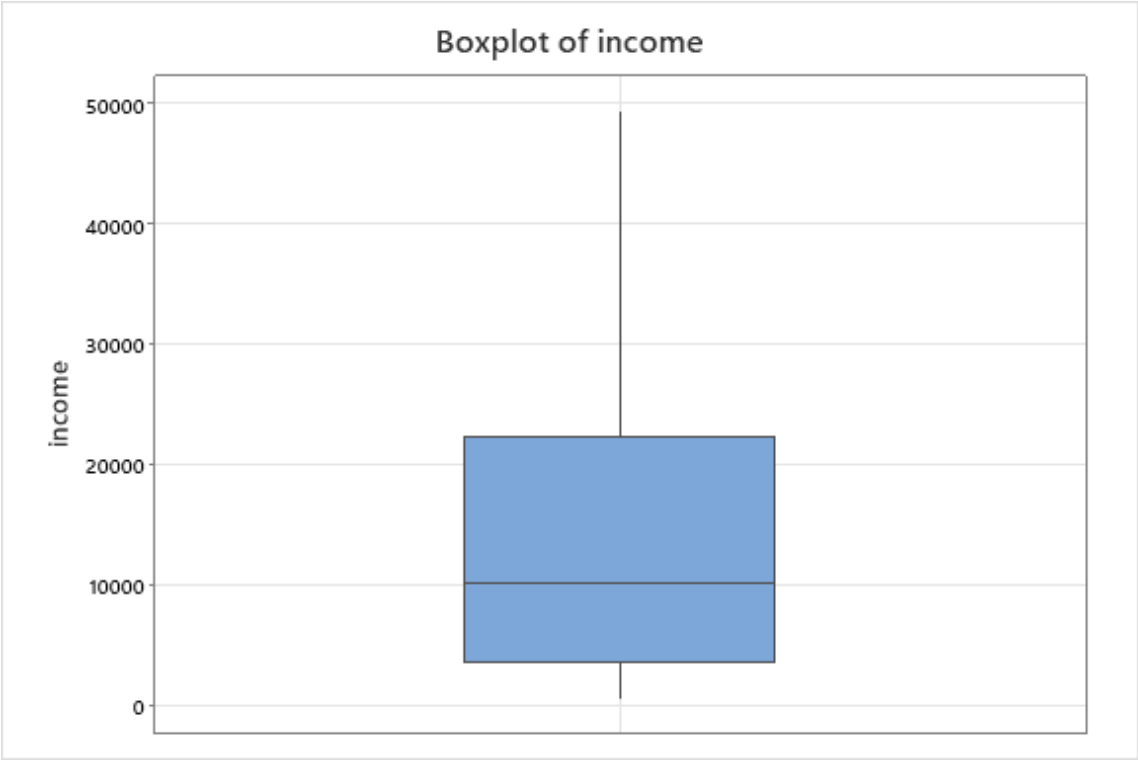

Boxplot of gdpp
gdpp
50000
40000
30000
20000
10000
0


Boxplot of health
health
12
10
8
6
4
2


Boxplot of imports
imports
100
80
60
40
20
0

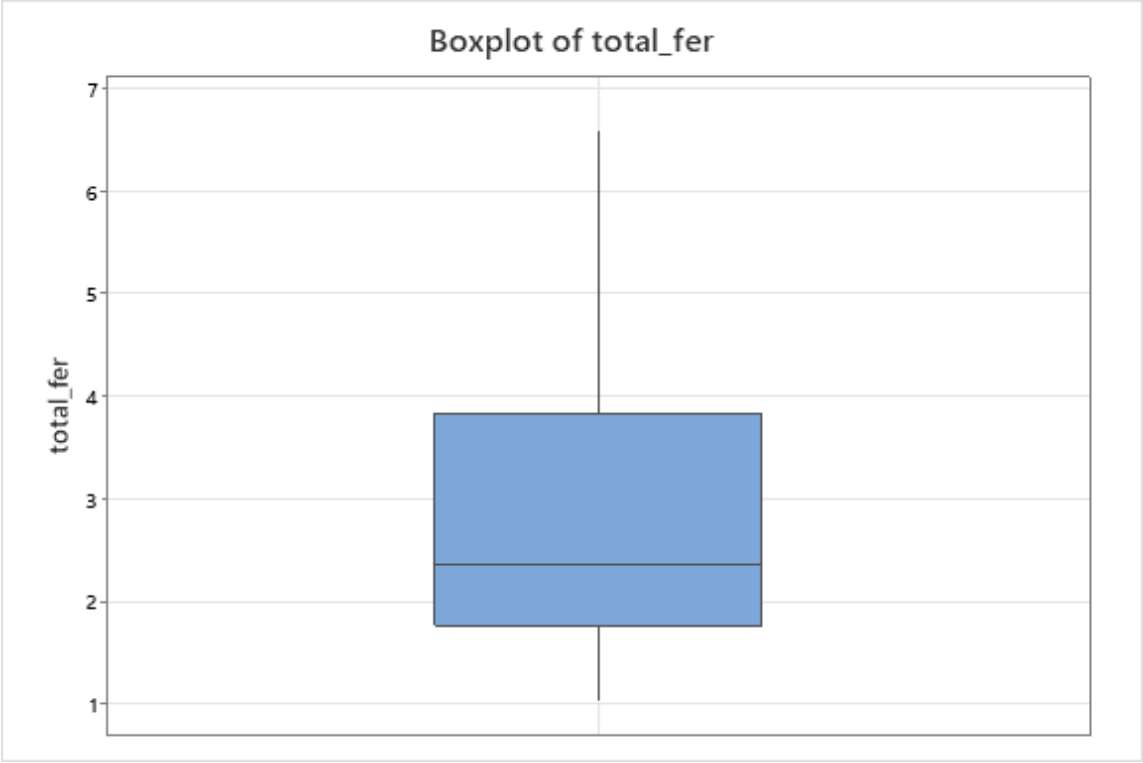

Boxplot of exports
exports
100
80
60
40
20
0

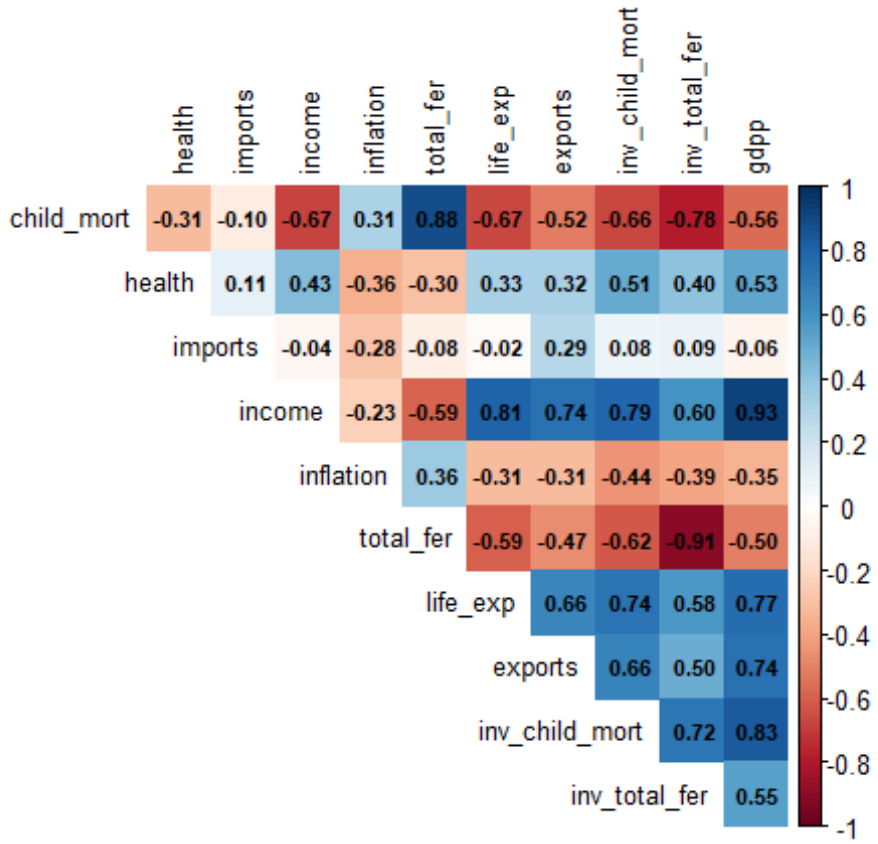

Boxplot of child_mort
child_mort
120
100
80
60
40
20
0


Boxplot of income
income
50000
40000
30000
20000
10000
0

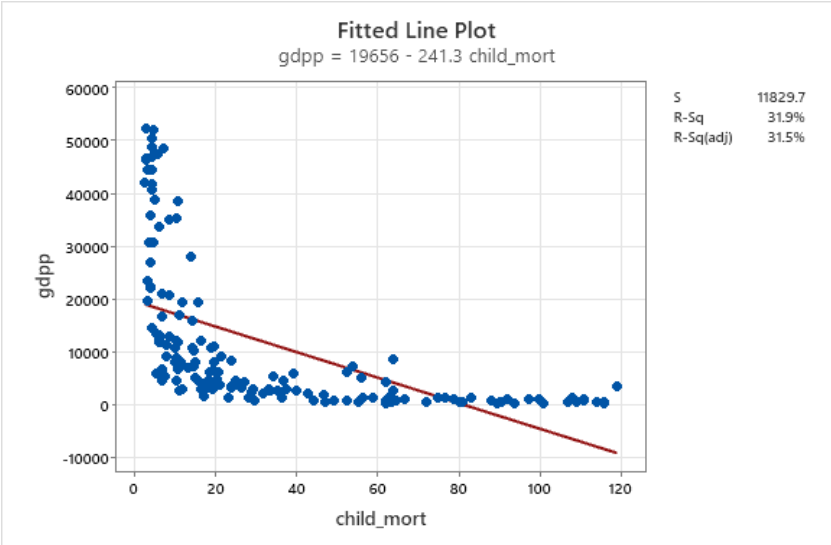

Boxplot of inflation
inflation
25
20
15
10
5
0
-5

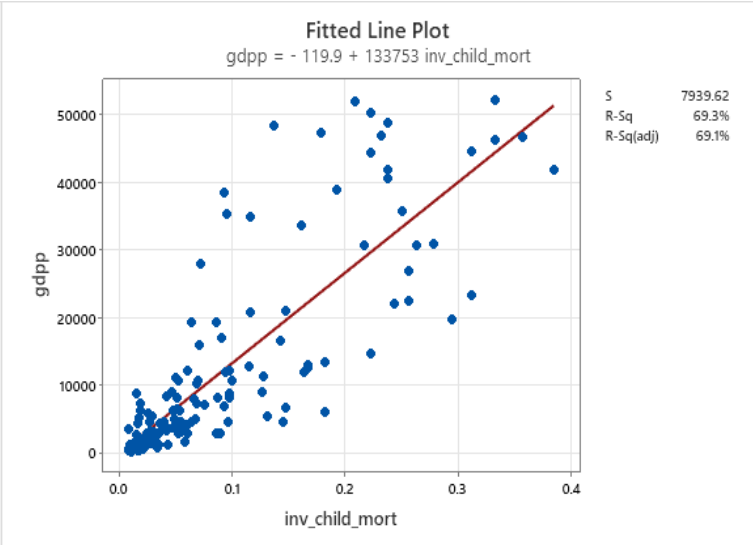

Boxplot of life_expec
life_expec
85
80
75
70
65
60
55
50

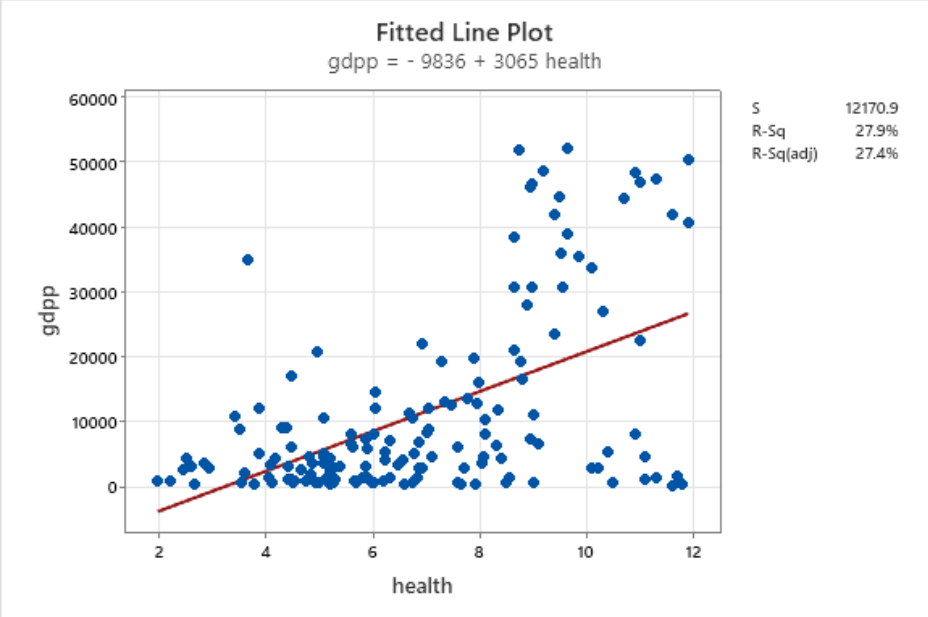

Boxplot of total_fer
total_fer
7
6
5
4
3
2
1

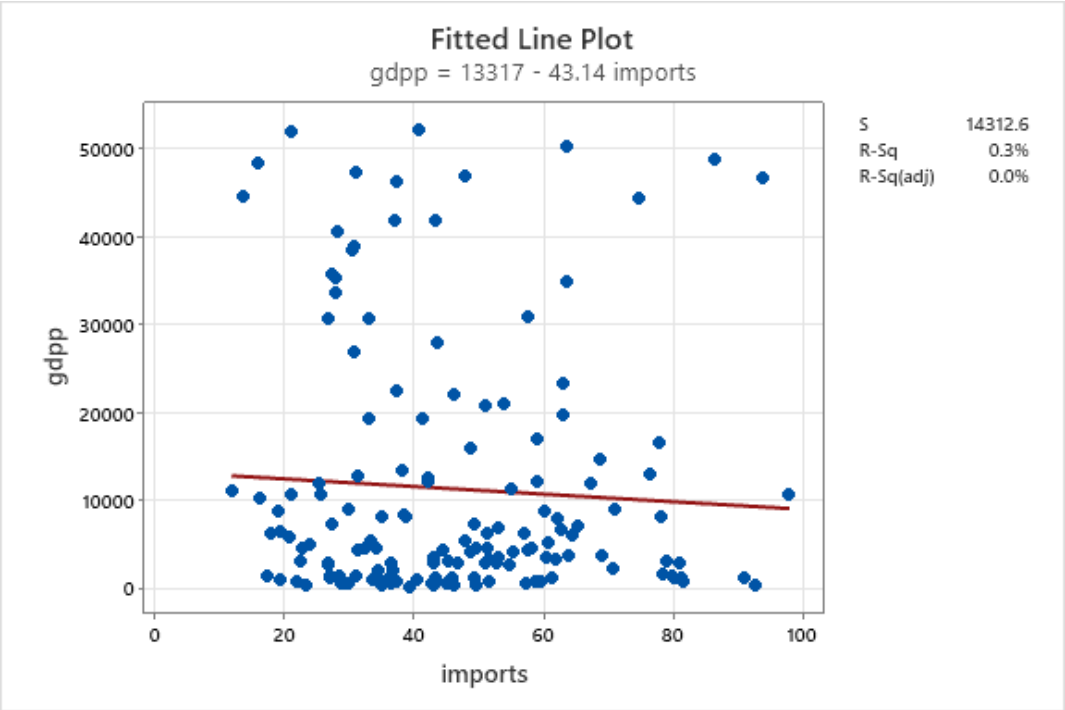

health
imports
income
inflation
total_fer
life_exp
exports
inv_child_mort
inv_total_fer
gdpp
child_mort -0.31 -0.10 -0.67 0.31 0.88 -0.67 -0.52 -0.66 -0.78 -0.56
health 0.11 0.43 -0.36 -0.30 0.33 0.32 0.51 0.40 0.53
imports -0.04 -0.28 -0.08 -0.02 0.29 0.08 0.09 -0.06
income -0.23 -0.59 0.81 0.74 0.79 0.60 0.93
inflation 0.36 -0.31 -0.31 -0.44 -0.39 -0.35
total_fer -0.59 -0.47 -0.62 -0.91 -0.50
life_exp 0.66 0.74 0.58 0.77
exports 0.66 0.50 0.74
inv_child_mort 0.72 0.83
inv_total_fer 0.55
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

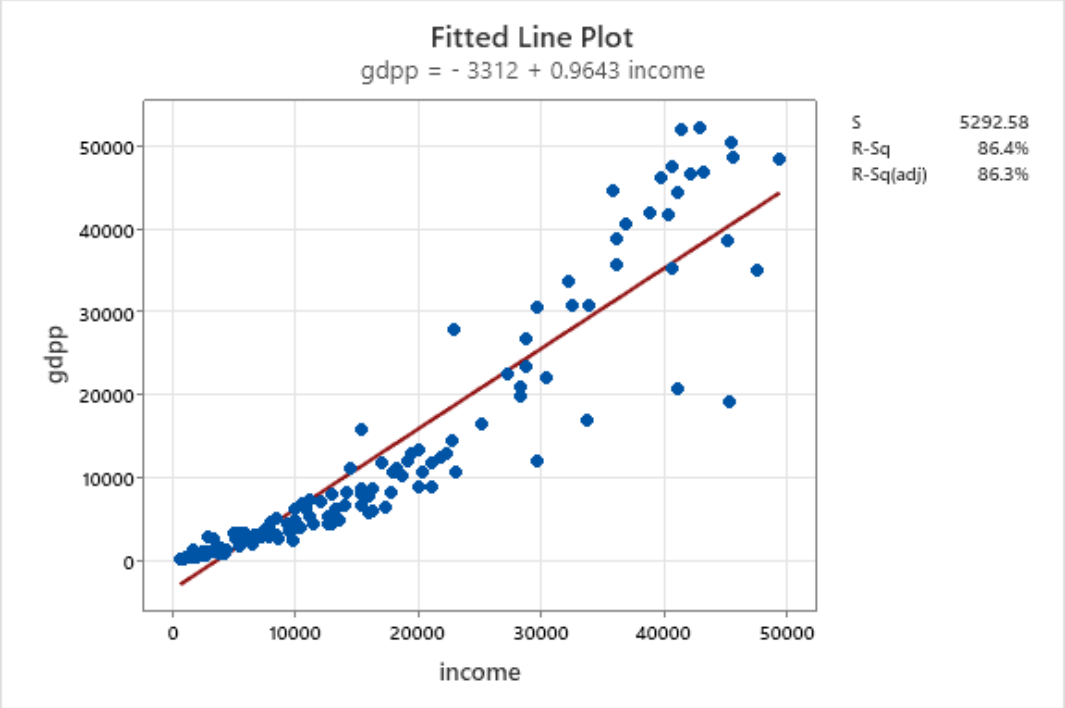
Regression lines

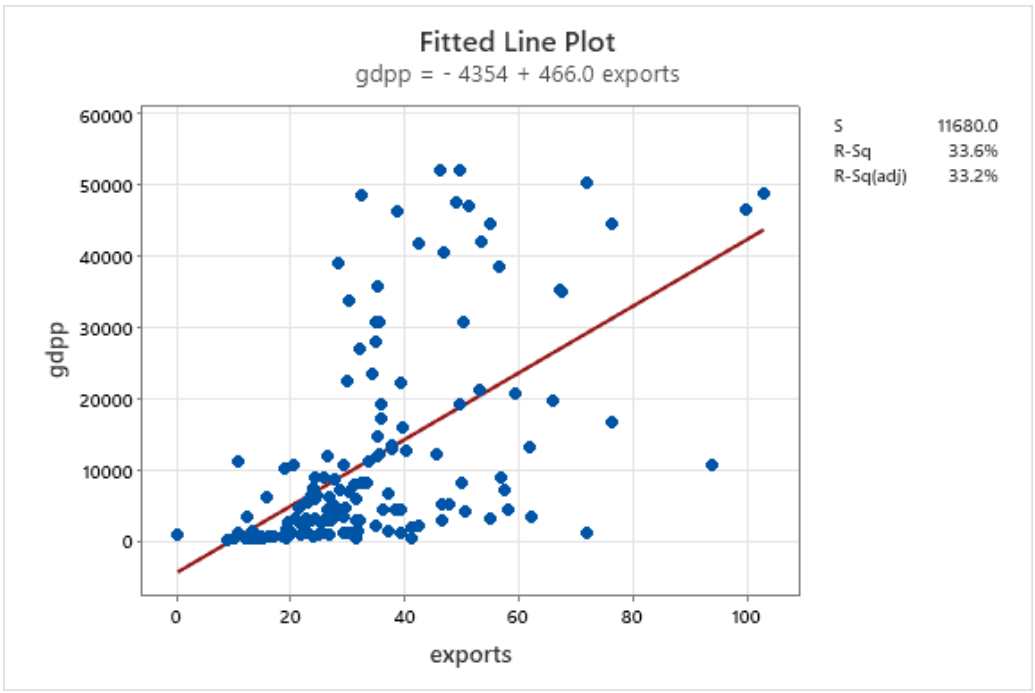


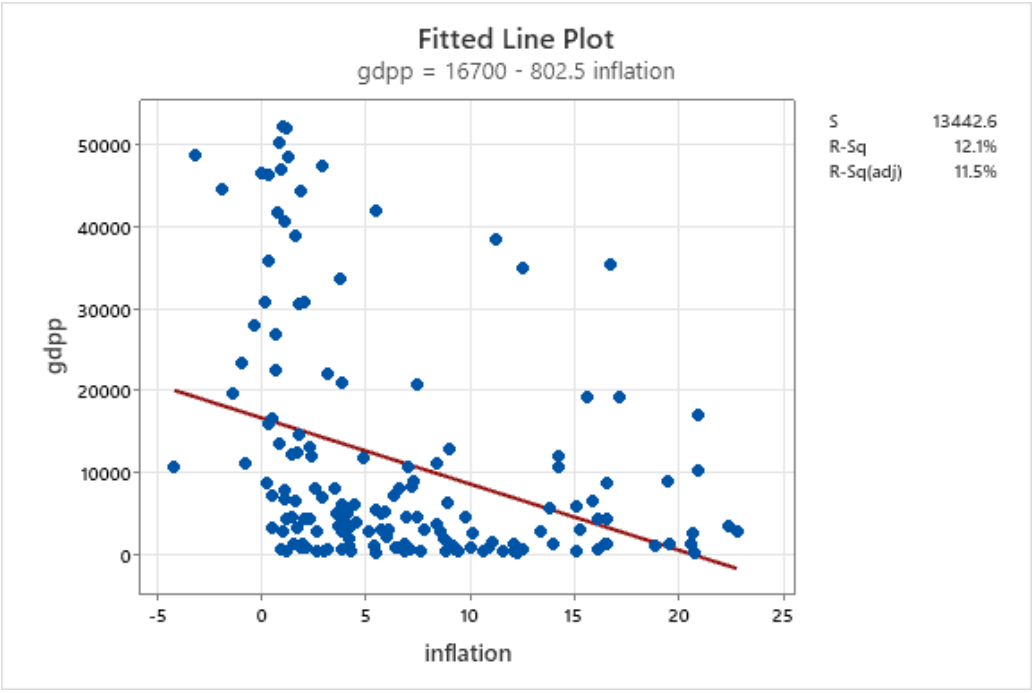


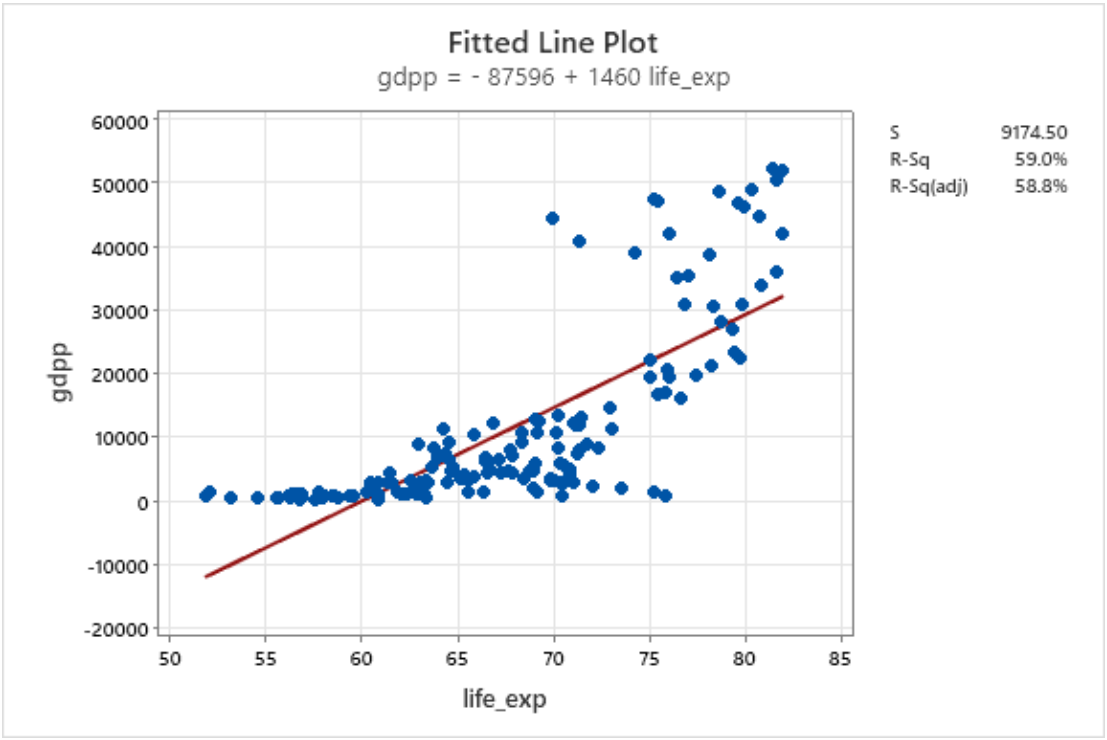


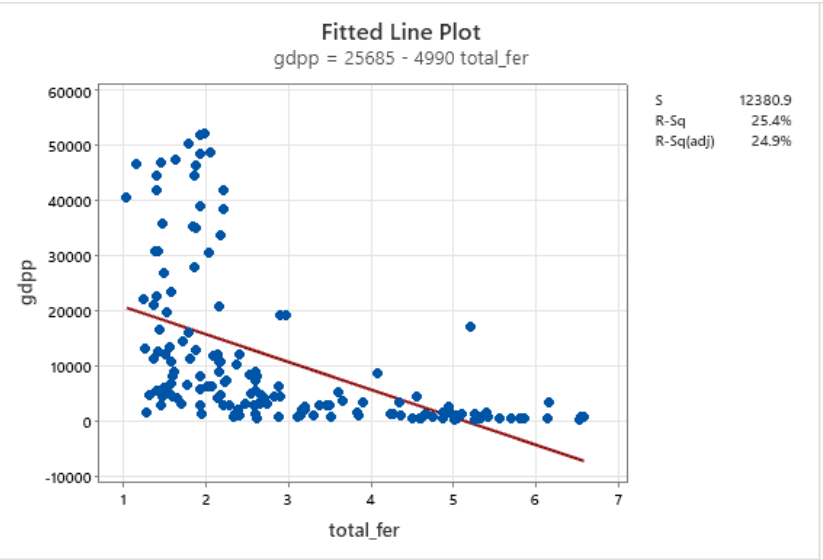

Fitted Line Plot
gdpp = 13317 - 43.14 imports
S 14312.6
R-Sq 0.3%
R-Sq(adj) 0.0%
gdpp
imports
0
10000
20000
30000
40000
50000
0
20
40
60
80
100

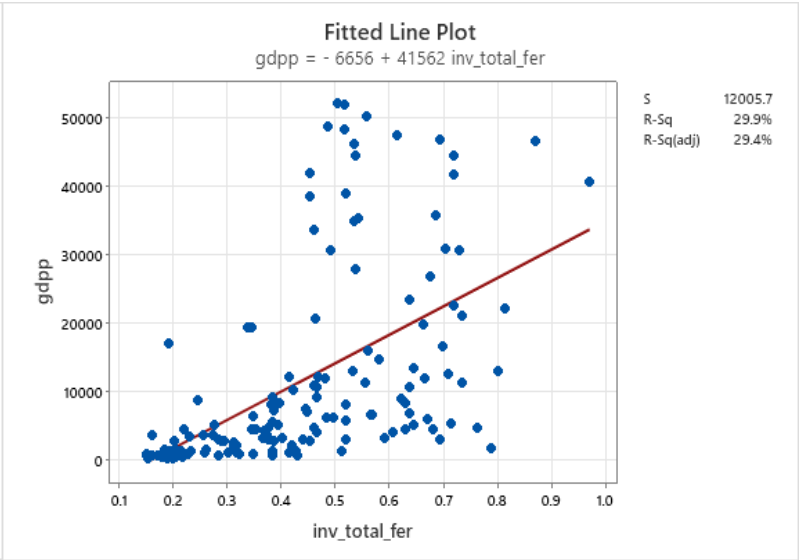

Fitted Line Plot
gdpp = - 3312 + 0.9643 income
S 5292.58
R-Sq 86.4%
R-Sq(adj) 86.3%
gdpp
income
0
10000
20000
30000
40000
50000
0
10000
20000
30000
40000
50000


Fitted Line Plot
gdpp = - 4354 + 466.0 exports
S 11680.0
R-Sq 33.6%
R-Sq(adj) 33.2%
gdpp
exports
60000
50000
40000
30000
20000
10000
0
0
20
40
60
80
100


Fitted Line Plot
gdpp = 16700 - 802.5 inflation
S 13442.6
R-Sq 12.1%
R-Sq(adj) 11.5%
gdpp
inflation
50000
40000
30000
20000
10000
0
-5
0
5
10
15
20
25


Fitted Line Plot
gdpp = - 87596 + 1460 life_exp
S 9174.50
R-Sq 59.0%
R-Sq(adj) 58.8%
gdpp
life_exp
60000
50000
40000
30000
20000
10000
0
-10000
-20000
50
55
60
65
70
75
80
85


Fitted Line Plot
gdpp = 25685 - 4990 total_fer
S 12380.9
R-Sq 25.4%
R-Sq(adj) 24.9%
gdpp
total_fer
60000
50000
40000
30000
20000
10000
0
-10000
1
2
3
4
5
6
7


Fitted Line Plot
gdpp = - 6656 + 41562 inv_total_fer
S 12005.7
R-Sq 29.9%
R-Sq(adj) 29.4%
gdpp
inv_total_fer
50000
40000
30000
20000
10000
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

```
> correlation_matrix
            child_mort     exports      health      imports      income  inflation
child_mort  1.0000000 -0.45351479 -0.24360333 -0.102382796 -0.67000134  0.3118306
exports    -0.4535148  1.00000000  0.07157109  0.409491760  0.61773714 -0.2339355
health     -0.2436033  0.07157109  1.00000000  0.108255422  0.26685635 -0.4034901
imports    -0.1023828  0.40949176  0.10825542  1.000000000 -0.04302752 -0.2838101
income     -0.6700013  0.61773714  0.26685635 -0.043027522  1.00000000 -0.2335321
inflation   0.3118306 -0.23393548 -0.40349013 -0.283810141 -0.23353211  1.0000000
life_expec -0.8679997  0.48454649  0.25573552 -0.006680413  0.71479145 -0.3582369
total_fer   0.8803261 -0.40380781 -0.24650593 -0.083395675 -0.58611314  0.3645860
gdpp       -0.5648658  0.57982713  0.39652525 -0.057023910  0.92935628 -0.3474725
             life_expec   total_fer        gdpp
child_mort -0.867999666  0.88032606 -0.56486578
exports     0.484546492 -0.40380781  0.57982713
health      0.255735518 -0.24650593  0.39652525
imports    -0.006680413 -0.08339567 -0.05702391
income      0.714791452 -0.58611314  0.92935628
inflation  -0.358236929  0.36458596 -0.34747247
life_expec  1.000000000 -0.82949601  0.66469654
total_fer  -0.829496009  1.00000000 -0.50412884
gdpp        0.664696543 -0.50412884  1.00000000
```

VIF values

```
> print(vif_values)

child_mort    exports     health    imports     income  inflation life_expec  total_fer
  6.730731   2.546738   1.308217   1.758531   3.218024   1.462514   5.598464   5.010491

> vif_values[vif_values > 5]

child_mort life_expec  total_fer
  6.730731   5.598464   5.010491
```

Basic Statistics

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
GDPP	154	0	11334	1151	14289	231	1340	4670	13200	52100

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
child mort	154	0	34.49	2.70	33.45	2.60	8.42	19.15	56.08	119.00

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
exports	154	0	33.67	1.43	17.78	0.11	22.50	29.95	40.35	103.00

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
health	154	0	6.699	0.202	2.506	1.970	4.855	6.270	8.512	11.900

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
imports	154	0	45.98	1.52	18.89	11.80	30.87	43.30	58.90	98.00

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
income	154	0	15189	1110	13772	609	3633	10180	22400	49400

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
inflation	154	0	6.686	0.499	6.187	-4.210	1.705	5.025	10.225	22.800

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
life_expec	154	0	70.927	0.641	7.951	52.000	65.375	73.150	76.725	82.800

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
total_fer	154	0	2.876	0.116	1.443	1.030	1.777	2.375	3.842	6.590